Scenario:
My client, a financial advisor, and her team have numerous subscriptions. Her job alone as financial advisor requires subscriptions to over 15 online products, each with varying uses and payment plans. These subscriptions are often forgotten or unused, leading to a large waste of time and money.
My client wants an automated system to track ongoing subscriptions, notify them of plan payments, and identify unused services. She specifically requested tools to analyse and breakdown spending, and the ability to choose from multiple payment plans per subscription. I proposed using a shared table to manage and add subscriptions with name, category, and payment plans. To identify unused services, we agreed to integrate screen time data for installed services, which could display in analytics and also sort and search functionality. They would also receive notifications for any upcoming payments and unused subscriptions. The client agreed on a website, and set a deadline to be when they would go over their expenses.

Rationale:
Product
I believe this is an appropriate solution as it provides reminders to prevent overlooked subscriptions, displays statistics for a clear breakdown, and integrates screen time tracking to provide empirical data on an app's value/usage. As well, a website is far more reliable and accessible than previous methods, which were manual and losable. The clients did not want to use available apps as they were tied to online banking services and were not integrated to provide screen time measurements.

Technology
Website was chosen to accommodate the various different devices and platforms of users. For the frontend, Vue was chosen as a Javascript web framework due to its rich ecosystem of tools, allowing for faster development and testing. Furthermore, Vue provides built-in reactivity, which allows for a more responsive and interactive user interface necessary for the various user inputs. Python's Django framework was chosen to develop the backend, as it provided many built in functions such as authentication, form handling, and databases. This would allow for faster iteration and time to develop more complex features such as analytics. Furthermore, Python has an extensive amount of libraries and documentation, particularly the NumPy library, which would be key to handling large amounts of screen time data. A SQLite database was used as it provided structured access, especially for relational data. This also provided flexibility in querying data and performing complex operations, which would be crucial for providing analytics. The Rescuetime API was chosen to track screen time as it provided a convenient, cloud-based solution which could integrate with existing apps. Being stored in the cloud, the website could gain consistent and reliable access to screen time data. The OneSignal API was chosen to send notifications as it has a simple and easily integratable Web SDK. It allowed for push notifications and ensured the client could receive notifications even when the app is not open.

Success Criteria:

1. Provide a secure register and log-in, and opt-in to stay logged across sessions.

2. All users are able to access, edit, and add to a shared subscriptions list, each with custom names, categories, and payment plans.

3. Users can add payment plans from the shared subscription list to a personal list, where payment dates and screen time are tracked and managed.

4. Integrates with screen-time tracking to provide usage data for installed services. Unused subscriptions will be flagged based on these metrics.

5. Users can set up notifications for upcoming payments and unused subscriptions, based on user-defined parameters like advance period and usage threshold.

6. Users can search for subscriptions by name and apply filters to sort by category, payment period, cost, and time usage.

7. Users can set a budget, apply it to specific categories, and receive recommendations to adjust subscriptions to stay within budget.

8. Users can view detailed analytics, such as spending per payment period, category breakdowns for usage and cost, and expensive subscriptions.